	SVKM's NMIMS **Mukesh Patel School of Technology Management & Engineering /** **School of Technology Management & Engineering**	
B. Tech CS	**Lab and Workbook**	**Academic Year- 2026-27**
Year:-Third	**Subject:-** Drone Technology	**Semester: - Fifth**

Experiment: 6

(PART A: TO BE REFERRED BY STUDENTS)

Aim: To find the specification and hardware components of drone.

- 1045 Propeller 10in 10x4.5 For Drone
- FlySky FS-i6 2.4G 6CH AFHDS Transmitter With FS-iA6B Receiver
- APM Pixhawk Power Module with XT60
- Nylon Strap Belt for RC Lipo Battery
- 3D printed Shock Absorber Anti-vibration Set for APM Pixhawk

- To assemble the hardware components for drone.

Pre-requisite: knowledge of components of drone

Learning Outcomes: The learner would be able to

1. **Understanding the purpose and function of each hardware component in a drone.**
2. **Developing the ability to assemble a drone by integrating various electronic mechanical & components.**
3. **Hands-on practice in building, configuring a drone system.**

Theory:

Drones are complex systems composed of a wide range of components, from basic mechanical parts like propellers and frames to advanced electronic systems such as flight controllers and power management modules. Each component plays a critical role in ensuring the drone's functionality, stability, and performance during flight. To successfully design, assemble, and operate a drone, it is crucial to understand the theory and operation behind each of these components.

(Students must write –

Explain the following components with their technical specification and pin configurations

- 1045 Propeller 10in 10x4.5 For Drone
- Fly Sky FS-i6 2.4G 6CH AFHDS Transmitter With FS-iA6B Receiver
- APM Pixhawk Power Module with XT60
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1. 1045 Propeller (10 in × 4.5 in)

The propeller converts the rotational torque of the BLDC motor into thrust. The code "1045" means a diameter of 10 inches and a pitch of 4.5 inches (the theoretical distance the propeller advances in one revolution). A quadcopter needs two clockwise (CW) and two counter-clockwise (CCW) propellers so that the reaction torques of the motors cancel and the drone does not yaw uncontrollably.

Parameter	Specification
Size	10 in diameter × 4.5 in pitch (≈ 254 mm × 114 mm)
Material	Nylon / glass-fibre reinforced plastic (ABS)
Types	CW (normal) and CCW (reverse / "R") pair
Mounting	Central hub bore with adapter rings to fit the motor shaft; secured with a prop nut or prop-saver
Typical use	Medium quadcopters (F450 / F550 class) with ~900–1000 KV motors on 3S–4S LiPo
Pins / electrical	None – passive mechanical part

Pin configuration: a propeller is a purely mechanical component and has no electrical pins. The only "interface" is the hub hole, which must be matched to the motor shaft diameter using the supplied adapter rings, and the CW / CCW marking, which must match the motor rotation.

2. FlySky FS-i6 2.4 GHz 6-Channel AFHDS Transmitter with FS-iA6B Receiver

The transmitter (TX) is the pilot's hand-held remote. It sends stick and switch positions as radio signals to the receiver (RX) on the drone, which gives them to the flight controller. AFHDS (Automatic Frequency Hopping Digital System) hops across frequencies in the 2.4 GHz band to avoid interference.

Parameter	FS-i6 Transmitter	FS-iA6B Receiver
Channels	6 (expandable up to 10 using iBUS)	6 PWM outputs (CH1–CH6); PPM and iBUS outputs
Frequency / protocol	2.4 GHz ISM band, AFHDS 2A	2.4 GHz, AFHDS 2A (2-way)
Power supply	4 × 1.5 V AA cells (≈ 4.0–6.5 V)	4.0 – 6.5 V DC (powered from FC / BEC)
Range	≈ 500 m or more in open field	–
Antenna	Single external antenna	Dual antenna (diversity)
Display	128 × 64 LCD, 4 trim buttons, 3 switches + 2 VR knobs	Bind button + status LED
Telemetry	Receives sensor data back (battery	Sensor port (iBUS)

	voltage, RSSI)	

Pin configuration of FS-iA6B receiver:

Port / Pin	Function
CH1 – CH6 (3 pins each)	Signal (S) – Power (+) – Ground (–) servo-style PWM outputs: CH1 Roll (aileron), CH2 Pitch (elevator), CH3 Throttle, CH4 Yaw (rudder), CH5 / CH6 aux switches (flight mode, etc.)
PPM	Combined (summed) PPM signal of all channels on a single wire – connected to Pixhawk RC IN
i-BUS Servo (OUT)	Digital serial output of all channels (14 channels) – connected to a Pixhawk serial port
i-BUS Sensor (IN)	Telemetry sensor input (e.g. voltage sensor)
Bind button	Hold while powering on to bind the receiver with the transmitter

3. APM / Pixhawk Power Module with XT60 connector

The power module steps the high LiPo battery voltage down to a stable 5 V (BEC) to power the Pixhawk flight controller and its peripherals, and at the same time measures battery voltage and current. These analog readings are given to the autopilot, which uses them for battery monitoring and low-battery failsafe. The XT60 connector connects the battery to the module and to the power distribution lines of the ESCs.

Parameter	Specification (typical)
Input connector	XT60 (battery side), XT60 / bare wire output to PDB / ESCs
Input voltage	≈ 6 V – 18 V (2S–4S LiPo)
Max current sensing	≈ 90 A
BEC output	≈ 5.3 V, 3 A to the flight controller
Sensing	Analog voltage and current outputs (calibrated in Mission Planner: voltage multiplier ≈ 10.1, amps per volt ≈ 17)

Pin configuration of 6-pin DF13 cable (to Pixhawk POWER port):

Pin	Signal
1	+5 V (VCC)
2	+5 V (VCC)
3	CURRENT (analog current sense)
4	VOLTAGE (analog voltage sense)
5	GND
6	GND

4. Nylon Strap Belt for RC LiPo Battery

A reusable nylon hook-and-loop ("magic sticker") strap, about 20 mm × 270 mm, used to hold the LiPo battery firmly on the frame. A loose battery shifts the centre of gravity and can fall out in flight or during a crash, so the strap keeps it secure and also protects it from vibration. The surface has an anti-skid grip.

Parameter	Specification
Size	20 mm × 270 mm (pack of 5 straps)
Material	Nylon with hook-and-loop fastening, anti-slip lining
Property	Reusable, light weight, strong, anti-skid
Pins	None – mechanical accessory

5. 3D Printed Shock Absorber / Anti-Vibration Set for APM Pixhawk

The Pixhawk has sensitive IMU sensors (accelerometer and gyroscope). Vibrations from motors and propellers reach them through the frame as noise, which degrades attitude estimation and causes poor stability, "toilet-bowl" drift and bad altitude hold. The anti-vibration mount is a 3D printed plate/frame with soft rubber / silicone damper elements that mechanically isolates the flight controller from the frame so only clean motion is measured.

Parameter	Specification
Material	3D-printed plastic (PLA/ABS) mount with soft rubber/silicone dampers
Mounting	Controller sits on dampers; plate fixed to frame with screws
Function	Filters high-frequency vibration to the IMU
Pins	None – mechanical part

PART B

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Prog/Yr/Sem: B.Tech CE / Third Year / Fifth Sem	**Batch:** C1
Date of Experiment: 15/10/24	**Date of Submission:** 15/10/24

Task to be performed

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Task 1: Understand the purpose, function, and specifications of drone component such as propeller, power module, transmitter and receiver, shock absorber.

Task 2: Assemble the Hardware Components for the Drone

Task 1: Purpose, function and specifications of the components

Component	Purpose / Function	Key specification
1045 Propeller	Produces thrust and lift; CW/CCW pair cancels yaw torque	10 in dia., 4.5 in pitch
FS-i6 TX + FS-iA6B RX	Wireless pilot control link between pilot and flight controller	2.4 GHz AFHDS 2A, 6 channels
Power Module (XT60)	Powers Pixhawk with 5 V BEC; senses battery voltage & current	≈ 90 A, 5.3 V BEC, 6-pin output
Nylon strap	Secures LiPo battery on frame	20 × 270 mm hook & loop
Anti-vibration set	Isolates IMU from motor/prop vibration	3D printed mount + dampers

Task 2: Assembly of the drone hardware (procedure followed)

- Mount the four BLDC motors on the arm ends of the frame and tighten the screws; note the motor numbering for Quad-X.
- Solder / connect the ESCs to the power distribution board and connect each ESC's three motor wires to its motor.
- Fix the Pixhawk on the anti-vibration shock-absorber mount and mount it at the centre of the frame, arrow pointing to the front.
- Connect the power module between the LiPo battery (XT60) and the PDB; plug its 6-pin cable into the POWER port of the Pixhawk.
- Connect the ESC signal wires to MAIN OUT 1–4 of the Pixhawk (Motor 1 front-right CCW, 2 rear-left CCW, 3 front-left CW, 4 rear-right CW).
- Mount the FS-iA6B receiver, bind it to the FS-i6 transmitter and connect its PPM / i-BUS output to the RC input of the Pixhawk.
- Fit the propellers last (CW on CW motors, CCW on CCW motors) and secure the battery with the nylon strap.
- Do a final check of wiring, polarity, tightness and centre of gravity before powering up.

Safety: propellers are attached only after all configuration and motor tests are completed.

Observations and Learning:

- Each component has a specific role: propellers produce thrust, the transmitter/receiver pair gives control, the power module powers and monitors the battery, the strap secures the battery and the anti-vibration mount protects the IMU.
- Only the receiver and the power module have electrical pin configurations (servo-style S/+/– pins, PPM, i-BUS and a 6-pin 5 V/CURRENT/VOLTAGE/GND cable); propeller, strap and shock absorber are mechanical parts.
- Propeller direction (CW/CCW) must match the motor rotation, otherwise the drone will not lift off and may flip.
- The power module supplies a stable 5 V to the flight controller so it is not affected by changes in motor load, and gives battery voltage/current for failsafe.
- Mounting the flight controller on dampers reduces vibration noise in the sensors and improves stability.
- Proper wiring order, polarity checks and fitting propellers last are essential for safe assembly.

Conclusion:

The specifications, functions and pin configurations of the 1045 propeller, FlySky FS-i6 transmitter with FS-iA6B receiver, Pixhawk power module with XT60, nylon battery strap and anti-vibration set were studied, and the hardware was assembled into a drone system. Learning outcomes 1–3 were achieved: the purpose of each part was understood, the components were integrated correctly (motors, ESCs, power module, flight controller, receiver, propellers and battery) and hands-on practice in building a drone was gained.